

MSCS 634 – Advanced Big Data and Data Mining

Comprehensive Written Report

Gym Members Exercise Dataset

Project Deliverable 4

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1. Introduction

This project analyzes the Gym Members Exercise Dataset to identify meaningful relationships among demographic characteristics, physiological measurements, workout behaviors, and fitness outcomes. The work consolidates the previous project deliverables into one end-to-end data mining workflow that includes data preparation, exploratory data analysis (EDA), feature engineering, regression, classification, clustering, and association rule mining. The objective is not only to evaluate predictive models, but also to determine which techniques provide useful and interpretable insights into gym-member exercise patterns.

The analysis uses supervised and unsupervised methods for complementary purposes. Regression predicts a continuous outcome, `Calories_Burned`; classification evaluates whether `Workout_Type` can be predicted from the available attributes; clustering searches for naturally occurring member profiles; and association rule mining identifies recurring combinations of workout and fitness characteristics.

2. Dataset and Rationale

The Gym Members Exercise Dataset contains 973 observations and 15 original variables. Each row represents one gym-member observation, and the variables include demographic, physiological, body-composition, and workout-related information. The dataset was chosen because its combination of numerical and categorical variables supports several data mining techniques within a single, coherent fitness domain.

Category	Variables
Demographic	Age, Gender
Body measurements	Weight (kg), Height (m), BMI, Fat_Percentage
Heart rate	Max_BPM, Avg_BPM, Resting_BPM
Workout behavior	Workout_Type, Session_Duration (hours), Workout_Frequency (days/week), Experience_Level
Fitness / activity	Calories_Burned, Water_Intake (liters)

This dataset is particularly suitable for the project because it allows continuous prediction, multiclass classification, behavioral segmentation, and frequent-pattern discovery. It also provides an opportunity to compare where predictive modeling works well and where the available variables are insufficient.

3. Data Preparation and Preprocessing

3.1 Data Quality Checks

The dataset was inspected for missing values, duplicate rows, inconsistent categorical labels, data-type issues, and potentially noisy observations. All 15 columns contained 973 non-null values, so no missing-value imputation was required. The duplicate check found zero duplicate records. Gender and `Workout_Type` values were consistently formatted, and the numerical and categorical columns were stored using appropriate data types.

Boxplots were used to inspect numerical features for unusual values. Potentially extreme values were reviewed rather than automatically removed. The notebook retained observations that appeared plausible within the context of the dataset, avoiding unnecessary loss of information.

3.2 Feature Engineering

Feature engineering was used primarily for the regression analysis to create predictors that better represent exercise intensity, training volume, body composition, and hydration. The engineered features were:

- Heart_Rate_Increase
- Heart_Rate_Intensity
- Workout_Load
- Weekly_Workout_Hours
- Lean_Body_Mass
- Fat_Mass
- Water_Per_Kg
- BMI_Category

Importantly, the target variable `Calories_Burned` was not used to create engineered predictors. This prevented target leakage and preserved the validity of the regression evaluation. Categorical variables were encoded and numerical features were scaled where required by the modeling technique.

4. Exploratory Data Analysis

EDA was used to understand variable distributions and relationships before model development.

Histograms showed generally reasonable numerical distributions without major gaps or obvious data-quality problems. The correlation analysis identified `Session_Duration` (hours) and `Calories_Burned` as the strongest relationship, with a correlation of approximately 0.91. The scatter plot for these variables also showed a strong positive linear pattern: longer sessions were generally associated with higher calorie expenditure.

Comparisons across `Workout_Type` showed relatively similar `Calories_Burned` distributions across Yoga, HIIT, Cardio, and Strength, although HIIT and Strength had slightly higher median values in the notebook visualization. This overlap foreshadowed the later classification result: workout type was difficult to distinguish using the available features.

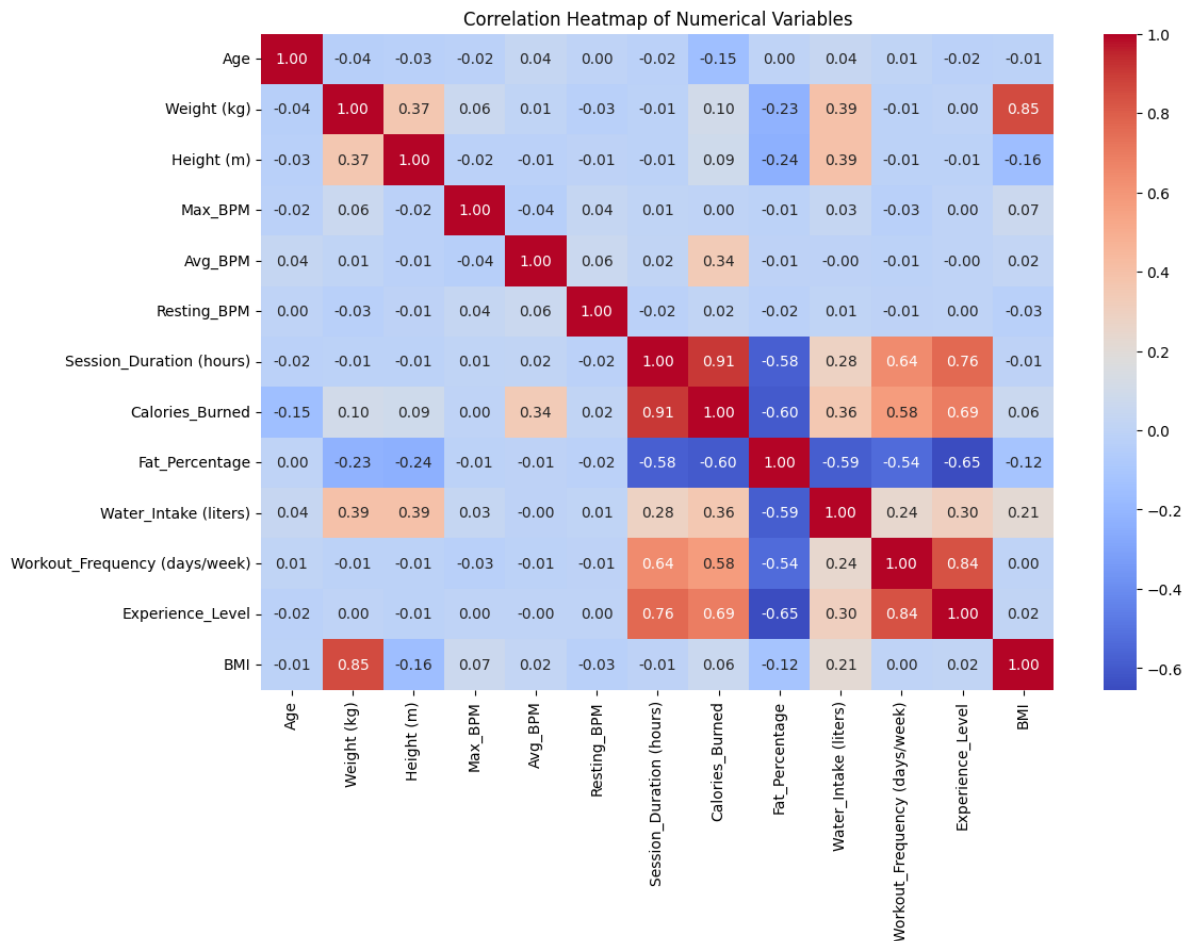


Figure 1. Correlation heatmap from the exploratory data analysis.

5. Regression Modeling and Evaluation

5.1 Modeling Approach

Regression analysis used Calories_Burned as the target variable. Multiple Linear Regression and Ridge Regression were evaluated using a holdout test set and 5-fold cross-validation. Model performance was measured using R-squared, mean squared error (MSE), root mean squared error (RMSE), and mean absolute error (MAE).

5.2 Results

Model	R ²	MSE	RMSE	MAE
Multiple Linear Regression	0.9854	1215.2923	34.8610	26.8968
Ridge Regression	0.9854	1214.7987	34.8540	26.8977

Both models explained approximately 98.54% of the variation in calories burned on the holdout test set. Ridge Regression produced a marginally lower MSE and RMSE, while Multiple Linear Regression produced a marginally lower MAE. The differences were extremely small, indicating nearly identical predictive performance. In 5-fold cross-validation, both models achieved a mean R-squared of approximately 0.9843, supporting the stability of the regression results.

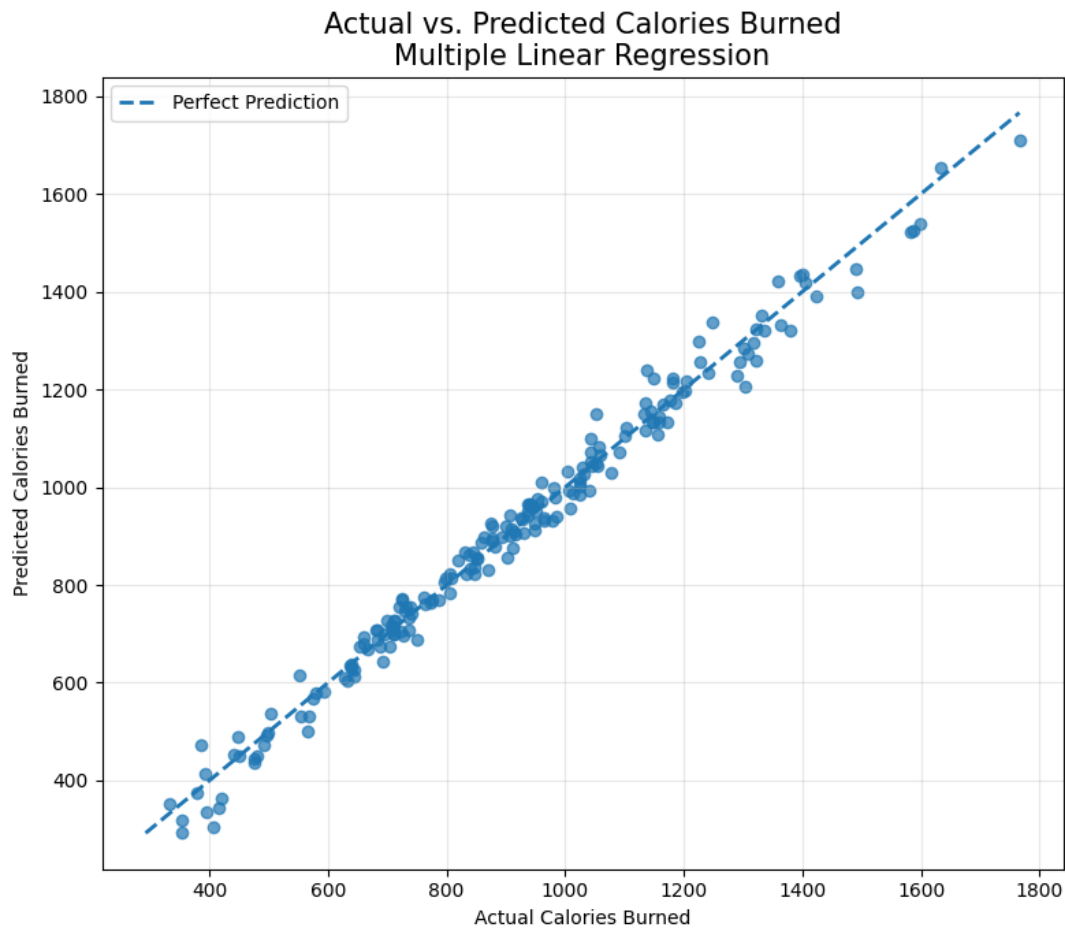


Figure 2. Actual versus predicted calories burned for Multiple Linear Regression.

The actual-versus-predicted visualization shows that most observations lie close to the perfect-prediction line. Residual analysis in the notebook also showed residuals centered around zero without a strong systematic pattern. Regression was therefore the strongest predictive component of the project.

6. Classification Modeling and Evaluation

6.1 Modeling Approach

The classification task used Workout_Type as the target, with four classes: Cardio, HIIT, Strength, and Yoga. Baseline Decision Tree and k-Nearest Neighbors (k-NN) classifiers were evaluated using accuracy, weighted F1 score, confusion matrices, and multiclass ROC analysis. The Decision Tree was also tuned using GridSearchCV.

6.2 Results

Model	Accuracy	Weighted F1
Decision Tree	26.15%	0.2597
k-NN	23.08%	0.2299
Tuned Decision Tree	21.03%	Lower than baseline

The Decision Tree slightly outperformed k-NN, but both models performed close to the expected random baseline for a relatively balanced four-class problem. Tuning did not improve generalization; the tuned Decision Tree accuracy decreased to 21.03%. These results indicate that demographic, physiological, and general workout measurements in this dataset do not provide clear class boundaries for predicting workout type.

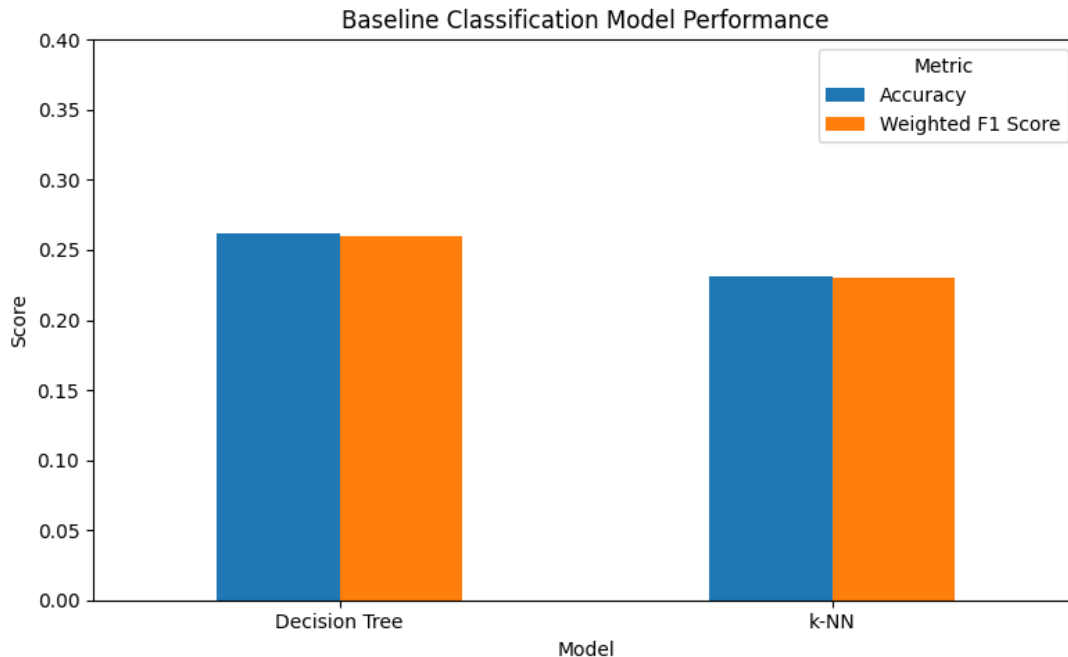


Figure 3. Baseline classification model performance.

This weak result is itself informative. Workout choice may depend on factors not represented in the dataset, such as personal goals, exercise preferences, injury history, motivation, equipment availability, or training environment. The project therefore avoids claiming that workout type can be reliably inferred from the current features.

7. Clustering Analysis

7.1 Modeling Approach

K-Means clustering was used to identify naturally occurring member profiles from standardized physiological and workout-related features. Candidate values of k from 2 through 10 were evaluated using inertia and silhouette score. The best-performing solution by silhouette score used $k = 2$.

7.2 Results and Interpretation

The final two-cluster solution achieved a silhouette score of approximately 0.231. Cluster 0 contained 198 members and represented a smaller, more experienced and active profile. These members had longer sessions, higher workout frequency, greater calorie expenditure, higher water intake, and substantially lower body-fat percentages. Cluster 1 contained 775 members and represented the majority profile, with shorter sessions, lower frequency, lower experience, lower calorie expenditure, and higher body-fat percentages.

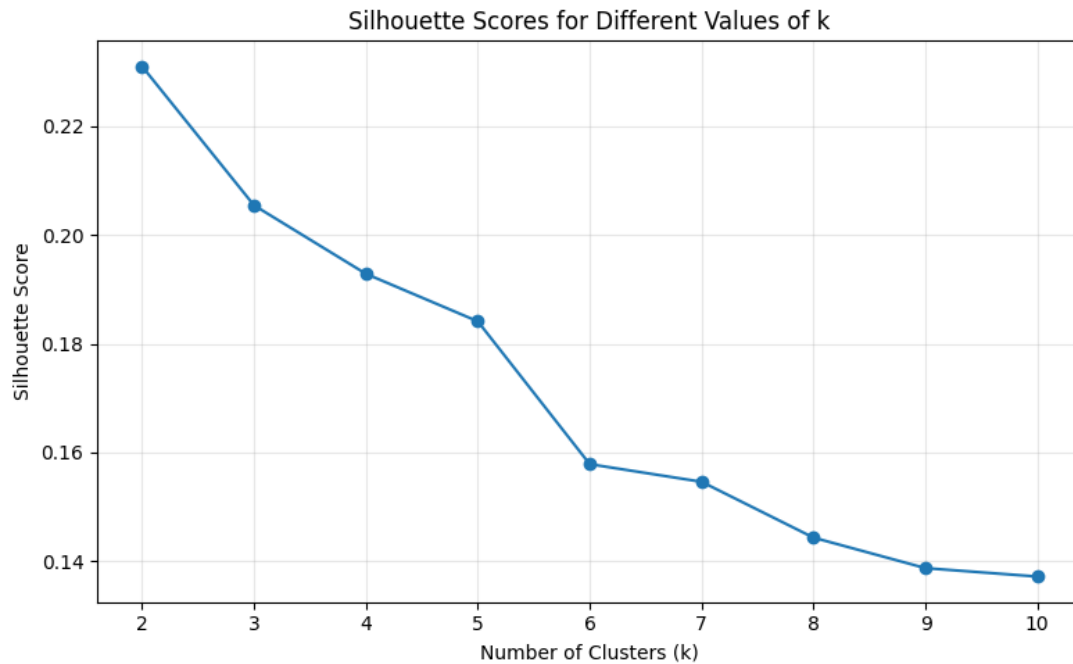


Figure 4. Silhouette scores for candidate K-Means solutions.

Age, weight, BMI, and heart-rate measurements were comparatively similar between the two clusters, suggesting that exercise behavior, experience, and body composition contributed more to the practical distinction. A PCA visualization captured approximately 48.16% of total variance in its first two components. Because the silhouette score was low, the clusters should be interpreted as broad, overlapping behavioral profiles rather than sharply separated populations.

8. Association Rule Mining

8.1 Approach

Association rule mining converted selected continuous and categorical characteristics into interpretable transaction-style categories and evaluated frequent patterns using support, confidence, and lift. The analysis focused on relationships involving workout frequency, experience, session duration, calorie expenditure, and body fat. Continuous variables were categorized to make the rules easier to interpret, while acknowledging that discretization reduces numerical detail.

8.2 Major Rules

Representative Rule	Support	Confidence	Lift
High workout frequency → Advanced experience	≈ 0.105	1.00	≈ 5.09
Advanced experience → Long workout session	—	—	≈ 3.08
Advanced experience → Low body fat	—	—	≈ 2.99
Advanced experience → High calorie expenditure	—	≈ 0.92	≈ 2.75

Short session $\leftrightarrow$ Low calorie expenditure	—	—	≈ 2.47
Long session $\rightarrow$ High calorie expenditure	—	—	≈ 2.27

The strongest representative rule was High workout frequency $\rightarrow$ Advanced experience, with confidence of 1.00 and lift of approximately 5.09. Advanced experience was also strongly associated with longer sessions, lower body fat, and higher calorie expenditure. The rule visualization showed many lift values above 1, indicating positive associations beyond what would be expected under independence.

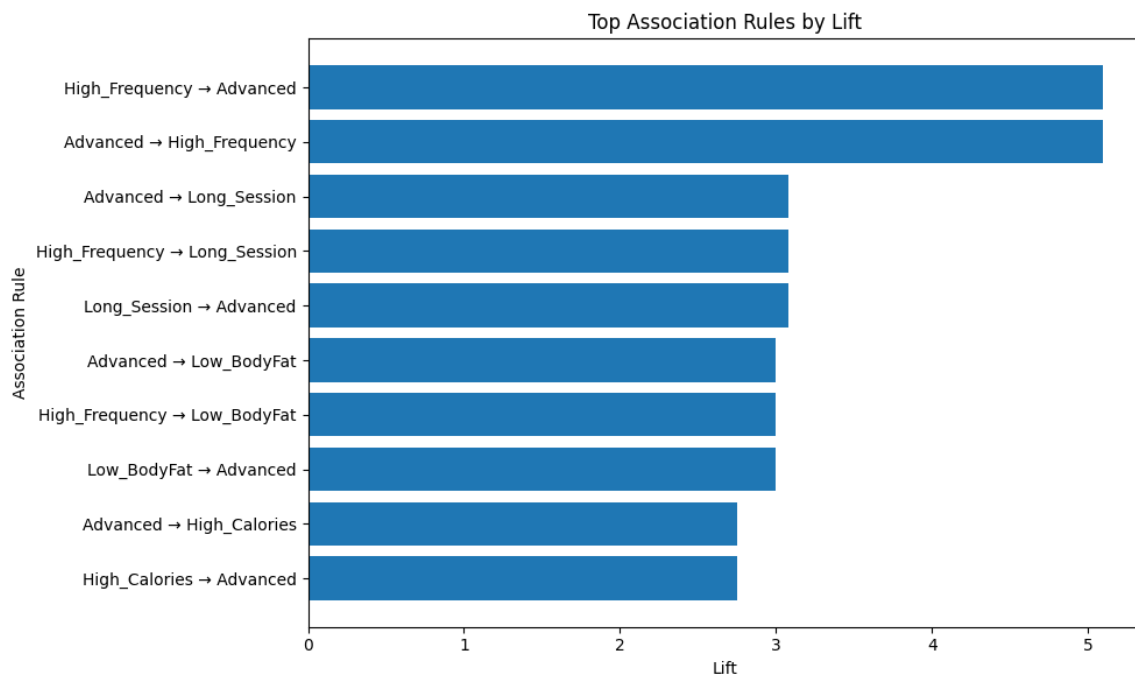


Figure 5. Confidence versus lift for association rules; marker size reflects support.

These rules reinforce the clustering findings: workout frequency, experience, duration, calorie expenditure, and body composition form a coherent behavioral pattern in this dataset. They remain associations, however, and do not establish causal relationships.

9. Integrated Insights

Taken together, the methods reveal a consistent distinction between what the dataset can predict well and what it can only describe. Calories_Burned can be predicted with very high accuracy from the available exercise and physiological features, while Workout_Type cannot be classified reliably. Unsupervised methods nevertheless reveal broad behavioral structure: more frequent and experienced exercisers tend to train longer, expend more calories, consume more water, and have lower body-fat percentages in this dataset.

- Session duration is one of the clearest variables associated with calorie expenditure.
- Workout type has substantial overlap across the available member characteristics and should not be inferred from these features alone.
- Exercise frequency, duration, and experience provide more useful segmentation signals than workout type.

- Association rules and clustering converge on similar patterns involving advanced experience, high frequency, longer sessions, higher calorie expenditure, and lower body fat.

10. Practical Recommendations

- Use broad behavioral profiles rather than workout-type predictions when segmenting members. The clustering results support differentiating members by activity level, frequency, duration, and experience.
- For less experienced or moderately active members, fitness programs can emphasize gradual progression in workout consistency and session duration rather than assuming a preferred workout category.
- Use the strong duration–calorie relationship to set realistic expectations about energy expenditure, while avoiding causal or individualized claims that exceed what the dataset supports.
- Do not use the classification models to assign or recommend workout types. Their near-chance performance is insufficient for reliable individual decisions.
- Future data collection should include fitness goals, exercise preferences, training history, injury history, motivation, nutrition, sleep, and equipment access if workout preference or personalized recommendations are important targets.

11. Ethical Considerations

11.1 Privacy

The dataset contains health- and fitness-related attributes such as body fat, BMI, heart rate, weight, and exercise behavior. Even though the project analyzes these variables at the dataset level and does not attempt to identify individuals, similar real-world systems should minimize collection of personally identifiable information, restrict access to sensitive health-related data, and use secure storage and appropriate consent practices.

11.2 Fairness and Bias

The dataset may not represent the broader gym-going population. Its demographic composition, collection process, and sampling frame may influence model outcomes, and the notebook does not establish that the sample is representative across age, gender, fitness level, geography, disability status, or other relevant groups. Therefore, the findings should not be generalized beyond the dataset without external validation. Gender and other demographic variables should also be handled carefully so that descriptive differences are not converted into stereotypes or unequal treatment.

11.3 Model Misuse and Interpretation

Several safeguards were applied through the analytical approach: target leakage was avoided in regression feature engineering; models were evaluated on held-out data and with cross-validation where applicable; weak classification performance and low cluster separation were reported rather than overstated; and association rules were explicitly interpreted as co-occurrence rather than causation. Clusters should not be treated as fixed labels for individuals, and the models should not be used to make medical, diagnostic, or high-stakes health decisions.

12. Limitations

- Classification performance remained close to chance level, showing that the available features were insufficient for reliable Workout_Type prediction.
- The K-Means silhouette score of approximately 0.231 indicates considerable overlap between the two clusters.
- The first two PCA components explained approximately 48.16% of variance, so the two-dimensional cluster visualization does not represent all information used by K-Means.
- Association rules identify co-occurrence rather than causation, and discretizing continuous variables into categories can affect which patterns are discovered.
- The dataset lacks potentially important variables such as nutrition, sleep, genetics, long-term training history, fitness goals, motivation, injuries, and health conditions.
- Very strong rules may reflect structural relationships specific to this dataset and require validation before generalization.

13. Conclusion

This project demonstrates the value of combining multiple data mining techniques to understand fitness data from different perspectives. EDA identified strong relationships such as the connection between workout duration and calories burned. Regression produced highly accurate predictions of Calories_Burned, while classification showed that Workout_Type could not be reliably predicted from the available variables. K-Means identified two broad behavioral profiles, and association rule mining reinforced relationships among workout frequency, experience, duration, calorie expenditure, and body composition.

The most important overall lesson is that strong descriptive or predictive relationships in one part of a dataset do not imply that every outcome can be modeled successfully. The project therefore combines useful findings with explicit limitations. Future work using larger, more diverse datasets and richer behavioral variables could improve generalizability and support more reliable, personalized fitness analysis.

14. References

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